

MASIRAH, Oman

WMO: 412880

Lat: 20.6714N

Lon: 58.8889E

Elev: 19

StdP: 101.10

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	19.3	6.1	5.8	24.4	8.2	6.8	24.1	11.0	20.4	10.1	21.0	4.5	300	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.7	37.0	24.3	35.4	24.7	34.2	25.1	28.2	32.2	27.7	31.6	27.2	31.2	5.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.4	26.7	22.4	30.1	26.1	21.6	29.8	91.4	32.2	88.7	31.8	86.7	31.4	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.2	11.1	10.2	DB	15.1	40.9	1.5	1.3	14.0	41.9	13.1	42.7	12.2	43.4	11.1	44.4	
			WB	11.1	29.2	1.5	1.6	10.1	30.4	9.2	31.3	8.4	32.2	7.3	33.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	22.9	23.6	26.0	29.3	31.3	30.5	28.0	27.1	27.6	28.3	26.7	24.4
	DBStd	2.83	1.31	1.49	1.77	1.79	1.30	1.44	1.17	0.92	1.02	1.26	1.12	1.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6254	401	380	496	579	659	613	558	531	528	566	499	445
	CDD18.3	3213	142	147	237	329	401	363	300	272	278	307	249	187
	CDH23.3	31327	491	702	1953	3872	5316	4603	2952	2296	2621	3263	2242	1018
	CDH26.7	11617	21	74	554	1712	2848	2263	955	600	767	1181	547	96
Wind	WSAvg	5.2	4.0	3.9	3.9	4.7	6.2	7.0	8.0	7.2	5.4	3.6	3.8	4.3
Precipitation	PrecAvg	31	3	5	7	8	0	2	1	1	0	1	3	3
	PrecMax	128	41	72	81	82	4	32	22	46	4	42	71	23
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	37	8	12	16	19	1	7	3	6	1	7	10	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	30.3	34.3	38.1	39.4	38.6	34.8	32.7	33.6	35.3	32.1	29.2
		MCWB	19.9	19.9	20.5	22.1	23.9	24.4	25.4	25.1	24.7	21.9	21.8	21.7
	2%	DB	27.1	28.5	32.0	36.0	37.2	36.1	33.0	31.1	32.0	33.6	30.7	28.2
		MCWB	20.5	20.6	21.1	22.1	24.5	25.9	25.5	25.0	24.8	22.5	23.1	21.6
	5%	DB	26.3	27.4	30.6	34.2	35.6	34.6	31.7	30.2	30.9	32.3	30.0	27.7
		MCWB	20.1	21.1	21.9	23.0	25.4	26.4	25.3	24.8	24.8	23.3	23.3	21.5
	10%	DB	25.7	26.7	29.5	32.8	34.2	33.3	30.6	29.5	30.1	31.1	29.3	27.1
		MCWB	19.8	21.0	22.2	23.7	26.0	26.6	25.1	24.6	24.6	24.0	23.0	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	24.4	25.7	28.4	29.3	29.2	27.2	26.5	26.2	27.1	26.0	24.7
		MCDB	26.0	27.0	29.2	31.8	33.7	33.3	31.3	30.1	30.7	30.0	29.6	27.3
	2%	WB	22.4	23.2	24.8	27.0	28.2	28.4	26.4	25.7	25.7	26.2	25.2	23.4
		MCDB	25.3	26.4	28.4	30.6	32.4	32.3	30.8	29.6	30.1	29.5	28.6	26.7
	5%	WB	21.6	22.6	24.3	26.3	27.6	27.8	25.9	25.2	25.3	25.8	24.7	22.7
		MCDB	24.9	25.8	27.9	30.1	31.8	31.6	30.2	29.1	29.5	29.1	28.2	26.2
	10%	WB	20.8	22.0	23.7	25.8	27.2	27.4	25.4	24.8	25.0	25.4	24.1	22.1
		MCDB	24.5	25.3	27.3	29.6	31.5	31.2	29.6	28.7	29.0	28.8	27.9	25.9
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.6	7.3	7.8	7.7	7.1	6.4	6.1	6.2	7.0	5.8	5.6
		MCDBR	6.4	7.4	8.7	9.6	9.4	8.9	7.9	6.8	7.0	8.4	6.4	6.0
		MCWBR	3.5	4.0	4.6	4.8	3.6	2.8	2.5	2.5	2.3	3.8	3.2	3.0
	5% WB	MCDBR	6.4	6.6	7.5	7.7	7.8	7.0	6.8	6.4	6.5	6.6	6.2	5.8
		MCWBR	3.9	4.1	4.1	4.2	3.0	2.6	2.5	2.6	2.4	3.0	3.3	3.2
Clear-Sky Solar Irradiance	taub	0.406	0.441	0.488	0.547	0.609	0.741	0.815	0.748	0.597	0.456	0.432	0.410	
	taud	2.261	2.135	1.991	1.874	1.764	1.545	1.464	1.556	1.798	2.117	2.198	2.248	
	Ebn at Noon	860	851	824	780	726	630	586	628	724	825	826	841	
	Edh at Noon	126	151	181	207	229	282	306	280	217	152	133	124	
All-Sky Solar Radiation	RadAvg	4.94	5.72	6.60	7.22	7.47	6.88	6.25	6.14	6.36	5.98	5.09	4.65	
	RadStd	0.12	0.15	0.21	0.17	0.09	0.46	0.48	0.30	0.22	0.13	0.16	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-0.63	-0.99	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	-0.63	-0.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page